

Follow up Study on Body Parameters of National Male Athletic Players through Maltron Bioscan.

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ABSTRACT

A follow up study of body composition has been conducted on 24 male National athletic players by using Maltron Bioscan 902-2 during year 2019-20. The data was taken in the month of October - November 2019 initially and follow up in the month of February - march 2020 (gap of 5-6 months). All players were instructed to keep all precautions taken before Bioscan test for accuracy of results. The result of study indicates there was no difference in body weight, body mass index, fat, total body water and resting metabolic rate as indicated by test of significant between initial and final test value of athletes. The study shows National level athletic players maintained their body parameters during six month period by keeping their exercise programme, nutrition and other factors affecting body composition constant. This trend of findings may be due to elite level players.

INTRODUCTION:

The change in body weight does not indicate the changes in fat mass and lean body mass. So body composition analysis require for athletes to know fat mass and lean body mass in their body weight. The parameters of body composition are influenced by age, exercise, nutrition, genetics, health, fitness and clinical factors etc. There are lot of factors responsible for changes in body composition. Increase in body weight without altering fat mass is beneficial for sports person because lean body mass increases in the body weight. Body composition analysis is useful for assessing nutritional status and predicting clinical

outcomes. Various techniques used to evaluate body compartments include anthropometry, isotope dilution, hydrometry, densitometry, dual-energy X-ray absorptiometry, magnetic resonance imaging, computerized tomography, neutron activation analysis and bioelectric impedance analysis (BIA). BIA is quick, simple, non-invasive, cheap, safe, can be done anywhere using a small handheld equipment, and thus suitable for large-scale studies. Modern nutritional surveys such as 3rd US National Health and Nutrition Examination Survey (NHANES III) have collected BIA data (Dumler & Kilates 2003). BIA is based on the principle that tissues are full

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of electrolyte-containing fluids and conduct an electric current. Current flow is affected by resistance (R , the opposition to flow through intra- and extracellular ionic solutions) and reactance (X_c , additional opposition from the capacitance effect of cell membranes and tissue interface). Impedance (Z), the frequency-dependant opposition of the flow of current, is a composite of R and X_c . The relationship between X_c and R in circuits is given by phase angle (PA), arctangent of X_c/R . Lower PA is associated with cell loss, whereas high PA indicates increased body cell mass. BIA studies are very limited in number. Jha et al. (2006) published a study on End Stage Renal Disease (ESRD) patient and 441 healthy Indians. Their work is landmark for comparison of results with healthy as well as diseased persons. Doddoli et al. (2015) reported a study effect of yoga training of 120 days on resting metabolic rate and body fat metabolism in context of body constitution (prakriti). They found Yoga training significantly increases the RMR and reduced the body fat.

Kumar, M. (2016) published the relationship between total body fat, RMR and Glycogen content among sportsmen. In 2020, Bandyopadhyaya et al. published relationship of body composition variables with selected physiological parameters of young sports person of different games. They conducted study on young sportsmen of

different games state participants trained in SAI eastern region from last 2-3 years. However, data on elite sportsmen like National athlete not available. Keeping this in mind few parameters of body composition and their follow-up has been considered for this study. The sample size of this research work is limited due to elite national athletes.

MATERIAL AND METHODS:

A follow up study has been conducted on 24 male National Athletic players of different events. Anthropometric measurement, height has been taken with standard anthropometer and body weight on electronic weighing machine. Age, height and weight were used for assessment of the body parameters on Maltron Bioscan 920 -2. Athletes were issued instruction not to consume large amount of water prior to testing, some medicines particularly diuretics may effect results. They were advised to take minimum 12 hrs rest before test as exercise significantly effects body water content. To avoid alcohol, tea, fizzy or energy drinks etc. at least 24 hrs prior to testing as most of these drinks are diuretics reducing the amount of water in body. Gap of meal 2 to 3 hrs and urinate 30 minute before testing. The subject lie flat keeping legs and hands apart and extremities not bent. The surface should not be conductive. The body should be properly

symmetrical and still when the data is being process. The cable set is identified with number. For full test ME4000 Cable was placed centrally directly below the 3rd knuckle of middle finger, the second Electrode on the crease of the wrist. The ME4000 Electrode on the foot was applied where the second and 3rd toe meet the foot, 2nd Electrode was placed at the crease of the ankle in line with shin bone. Full test was performed by entering age, sex, height, weight, population and athletic option into Bioscan. At the time of testing the screen display the left to right frequencies 5KH, 50KH, 100KH and 200KH from top to bottom. Selected parameters were noted for interpretation of results. Follow-up study was conducted in the month of October –November 2019. Follow up measurement were taken in the month of February- March 2020. Six month changes in the body parameters have been recorded. Mean, standard deviation were computed, test of significance (t-test) was used to find out differences in initial and final parameters of players.

RESULT AND DISCUSSION:

Table 1 shows the overall descriptive statistic of anthropometric parameters for National campers. As evident from Table, on an average body weight increased 1.28 kg in six month when all players grouped together. The mean age was 23.13 (± 3.44 Yr.) when test was

conducted first time. The increase in body weight was statistically non-significant. The Body mass index (BMI) has also been increased due to increase in body weight.

However there was no change in fat percentage, total body water (hydration status), resting metabolic rate (RMR) increase just 16 Kcal. These results indicate that there was more increase in lean body mass compare to fat in players due to increase in body weight. However, results of these parameters were non-significant. Thus, training played good role in maintaining body parameters.

As shown in Table 2 & 3 Body weight, BMI and fat increases slightly in 400m runners. However, results were statistically non-significant. The changes in body fat, fat free mass, total body water and resting metabolic rate (RMR) were also non-significant during follow-up the study (Table 3).

The body weight, BMI and fat mass also decreased in 800m runners whereas fat free mass, total body water and resting metabolic rate increased slightly. However, comparisons of these parameters were statistically non-significant. The sample size was very small to draw conclusions. In 1500m runner's body weight, BMI and Fat remains same during 1st and 2nd phase of study. Fat free mass, total body water and resting metabolic rate also remains same during follow-up (Table 3). Similar findings were observed in

3000m Steeple chase runners. The Javelin throw players also showed slight increase in these parameters but non-significantly during this follow-up study. In 400m Hurdle and shot put number of player was one each. So statistic not applied. Body fat increase 1% in 400m Hurdle and body weight, BMI and fat also increased in shot put player, respectively.

As shown in Table 4 the National athletic players of present study were older, taller, heavier and leaner in body fat than football, hockey, Table tennis and badminton players of Bandopadhy et al. (2020). The sample of different sports reported by Bandyopadhy et al. (2020) belongs to SAI eastern region of state level junior athletes having 2-3 year training. The hydration status (total body water) has been found almost similar in all grouped sports.

Table 5 shows the comparative values of present study players and control group of North India studied by Jha et al. (2006). The male and female subjects of control group were older, shorter and heavier for height than athletic players. Athletic players were leaner in fat and more hydrated than controls in term of total body water. This difference may be due to age, heredity, dietary pattern and activity of athletes.

Kumar, M (2016) published a study on relationship between total body fat, resting metabolic rate

glycogen content among 29 sportspersons of LNIPE Gwalior. The sportspersons age range was 17 to 25 year. Their mean Resting Metabolic Rate (RMR) was 1620.96 kcal (± 175.32 kcal), total body fat 13.94kg (± 6.08 kg) and glycogen 467.58 gm (± 63.27 gm). He reported correlation between RMR and body fat -0.774* whereas RMR and glycogen 0.928*. Correlation between body fat and glycogen was found to be -0.573*. All correlation values were significant in statistically terms.

Doddoli et al. (2015) published body composition study on 60 residential healthy subjects of both gender ranging in age 18 to 45 years of GS College of Yoga and cultural Synthesis, Kaivalyadhama, Lonavla. The study was on effect of yoga training on Resting Metabolic Rate and body fat metabolism in the context of body constitution (Prakriti) using Maltron Bioscan. The sample was divided into two groups Vatta-Pitta (N=43), Pitta-Kapha (N=12) and Vatta-Kapha (N=2 excluded from study). Yogic training was given 120 days. In both groups i.e. Vatta-Pitta and Pitta-kapha there was significantly improvement in Resting metabolic rate, Body fat percent and total body fat after 120 days of study. In case of Vatta-pitta the mean values RMR was 1347 kcal (± 161), body fat% 33.97 (± 4.98), body fat 18.36 kg (± 3.72) after 120 days RMR 1454 Kcal (± 178 Kcal), Body fat 18.10 % and total body fat 10.83kg (± 3.5). Similarly, Pitta- kapha showed

differences in RMR 1470 ± 160 kcal (final value 1581 ± 189 kcal), Body fat% 30.80 ± 7.14 & final value $17.56\% \pm 7.72$ and total body fat $18.80\text{kg} \pm 5.69$ final value (10.64 ± 5.0). Thus yogic training effects body composition parameters.

In conclusion, more data and lengthy time period requires to establish differences in body compositions of athletes between players of same sports as well as in different sports. The study provided individual as well as in group body composition differences among different events of athletic. The variations observed body composition parameters after follow-up of six month period was non-significant in athletic players. This may be due to elite level players having well maintained nutrition and exercise. Large sample size on control and national athletes will be requirement for establishing

differences of body composition due to exercise during lengthy period of training. For short term period major changes in body weight may change composition only. Thus findings of study were related with body weight changes in athletes.

Thus in general, all parameters remain same during 1st and 2nd phase of study. These finding indicate national level players well maintained their anthropometric variable during six month period by keeping their exercise programme, nutrition and other factors affecting body composition constant/optimal. Such results may be due to lesser sample size. These studies are required for longer period on large samples so that scientific findings regarding body parameters may be established. Such results will be utilized for future research planning on athletes.

Table : 1
Overall Descriptive Statistic of Anthropometric Parameters
for Athletic National Campers

Parameters	1 st Phase (N=24)		2 nd Phase (N=24)		t value
	Mean	SD	Mean	SD	
Age (yrs)	23.13	3.44	-	-	-
Height (cm)	177.00	6.33	-	-	-
Weight (kg)	71.15	17.43	72.33	18.48	0.22
BMI	22.54	4.27	22.90	4.52	0.28
Fat %	11.91	1.59	12.13	1.51	0.49
FFM %	88.08	1.59	87.89	1.52	0.42
TBW %	61.56	2.84	61.81	2.87	0.30
RMR (Kcal)	2049.21	286.82	2065.50	307.99	0.19

Table : 2

Event wise Distribution of Weight, BMI & Fat % of Athletic National Campers

Event		Weight		t value	BMI		t value	Fat %		t value
		1 st Phase	2 nd Phase		1 st Phase	2 nd Phase		1 st Phase	2 nd Phase	
400 m (N=9)	Mean	65.58	67.26	0.60	21.30	21.84	0.99	11.48	12.14	1.86
	SD	5.60	6.30		1.05	1.25		0.80	0.70	
800 m (N=2)	Mean	67.9	67.5	0.06	21.37	21.25	0.09	12.66	11.64	1.57
	SD	6.92	4.94		1.55	0.93		0.70	0.59	
1500 m (N=4)	Mean	58.2	58.5	0.06	19.80	19.90	0.07	11.35	11.15	0.21
	SD	6.62	6.41		2.11	2.01		1.53	1.15	
3000 m SC (N=2)	Mean	58.5	58	0.24	18.7	18.49	0.17	11.19	11.19	0.0
	SD	0.70	2.82		1.62	0.48		2.24	0.94	
Javelin Thrower (N=5)	Mean	86.5	87.9	0.38	26.42	26.84	0.21	12.46	12.60	0.12
	SD	6.61	4.97		3.30	3.00		1.66	1.84	
400 m H (N=1)	-	65.8	66	-	21.24	21.31	-	10.64	11.53	-
Shotput (N=1)	-	133.5	140	-	36.52	38.3	-	16.51	16.89	-

Table 3

Event wise Distribution of FFM %, TBW % & RMR of Athletic National Campers

Event		FFM%		t value	TBW%		t value	RMR Kcal		t value
		1 st Phase	2 nd Phase		1 st Phase	2 nd Phase		1 st Phase	2 nd Phase	
400 m (N=9)	Mean	88.51	87.85	1.86	61.42	61.23	0.22	1956.66	1969.11	0.21
	SD	0.80	0.70		2.15	1.48		116.77	128.78	
800 m (N=2)	Mean	87.34	88.36	1.48	60.29	61.02	1.29	1976.5	1986.5	0.15
	SD	0.70	0.59		0.40	0.69		84.14	38.89	
1500 m (N=4)	Mean	88.64	88.91	0.28	62.79	63.25	0.21	1829.5	1827.5	0.03
	SD	1.53	1.12		3.44	2.83		95.40	99.21	
3000 m SC (N=2)	Mean	88.81	88.80	0.01	61.83	62.04	0.12	1851	1840	0.13
	SD	2.24	0.94		2.33	0.14		62.22	103.23	
Javelin Thrower (N=5)	Mean	87.52	87.39	0.11	60.43	61.40	0.31	2314	2356	0.82
	SD	1.66	1.84		4.45	5.50		95.92	61.93	
400 m H (N=1)	-	89.36	88.47	-	62.02	60.92	-	1982	1975	-
Shotput (N=1)	-	83.49	83.11	-	64.97	65.15	-	3046	3132	-

Table : 4
Body composition values in different sports.

Parameter	Athletic (present study) N=24	Football* (N=97)	Hockey* (N=110)	Table tennis * (N=75)	Badminton * (N=63)
Age (Yr)	23.13 (± 3.44)	17.0 (± 1.87)	16.2 (± 1.87)	16.8(± 2.5)	17.3(± 2.97)
Height (cm)	177.0 (± 6.33)	166.5(± 1.87)	169.1(± 4.57)	165.0(± 5.51)	170.1(± 5.64)
Weight (kg)	71.15 (± 17.43)	57.2(± 5.35)	57.9(± 6.25)	56.9(± 10.41)	60.0(± 8.99)
BMI	22.54 (± 4.27)	20.6(± 1.51)	20.3(± 1.85)	20.8(± 3.23)	20.9(± 2.83)
Body fat (%)	11.91 (± 1.59)	12.9(± 4.06)	13.1(± 4.01)	15.7(± 5.36)	15.7(± 4.88)
TBW (%)	61.56 (± 2.84)	62.3(± 4.01)	62.4(± 3.93)	59.9(± 5.68)	61.6(± 4.22)

*Studied by Bandyopadhyaya et al. (2020)

Table 5
Body composition values of athletic players and control group

Parameter	Athletic (present study) N=24	Control* (N=277) Indian male	Control* (N=174) Indian Female
Age (Yr)	23.13 (± 3.44)	39.0 (± 12.0)	37.0 (± 12.0)
Height (cm)	177.0 (± 6.33)	170.4(± 6.8)	157.5(± 6.2)
Weight (kg)	71.15 (± 17.43)	70.9(± 13.0)	62.4(± 12.4)
BMI	22.54 (± 4.27)	24.4(± 4.1)	25.1 (± 5.0)
Body fat (%)	11.91 (± 1.59)	21.6(± 7.98)	32.0 (± 10.6)
TBW (%)	61.56 (± 2.84)	56.5(± 3.6)	51.1 (± 3.5)

*Jha et al. (2006)

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